MARIA ELLIS (ELLIE) WHITE

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Education

Marshall University, Huntington, WV
M.S. in Physics *May 2026*
B.S. in Physics (Honors) with a Minor in Mathematics, GPA: 4.00 *December 2021*

Research and Technical Work Experience

NSF Graduate Research Fellow *Fall 2025 – Present*

- Analyzed the velocity and acceleration behavior of known and simulated Solar System objects using data from the LSST DP0.3 simulated data release, the Minor Planet Center, and NASA's Small Body Database / JPL Horizons.
- Simulated ephemerides of objects experiencing a range of velocities and non-gravitational accelerations using the ASSIST Solar System simulation software.
- Modified the LSST Solar System survey simulation software Sorcha to enable the simulation of objects experiencing non-gravitational accelerations.
- Active member of the Legacy Survey of Space and Time (LSST) Solar System Science Collaboration (SSSC)
- Attended the 2026 LSST SSSC's First Data Sprint in Belfast, where I presented a lightning talk on my research.
- Contributed to two science papers: Chandler et al. 2025 (on Rubin's observations of interstellar object (ISO) 3I/ATLAS) and Davenport et al. 2025 (on ISOs as technosignature search targets).

Peer reviewer for multiple astronomical journals *Summer 2022-Present*

- Reviewed papers for the *Publications of the Astronomical Society of the Pacific* and *Advances in Astronomy* in the field of single-dish radio telescope pointing and metrology systems.

Green Bank Observatory (GBO), Scientific Data Analyst *February 2022 – October 2022*

- Worked on projects related to the performance of the telescope, including pointing model analysis and serving on the High Frequency Task Force
- Provided backup support on the Breakthrough Listen project
- Assisted with workshops to train scientists and students how to observe with the Green Bank Telescope
- Assisted with the organization of workshops and events onsite
- Served as an active member of the Employee Diversity Group
- Participated in outreach events onsite, including mentoring middle school student groups and assisting with a star party
- Gave tours of the Green Bank Telescope: safely guided visitors up the exposed structure at heights over 400 ft and explained the technical background and history of the telescope.

Breakthrough Listen (BL), Green Bank Telescope Observer *Jan. 2021-Present*

- Fully qualified to operate the GBT's BL backend.
- Serve on a rotation of observers that monitor GBT BL observations.
- Work with GBT operators to troubleshoot observing issues as they arise.

NASA WV Space Grant Consortium, Undergraduate Research Fellow *Aug. 2020-June 2021*

- Researched the development of improved pointing models for the GBT.
- Documented the GBT's pointing and metrology systems on the GBO wiki and internal memos.
- Planned and undertook observations of the Mars2020 spacecraft using the Breakthrough Listen (BL) backend on the GBT. Analyzed resulting data and wrote up the results as a Kaggle tutorial.
- Analyzed spectral occupancy statistics in BL data from the GBT.

Berkeley SETI Research Center, Research Experience for Undergraduates *June 2020-Aug. 2020*

- Developed a GNU Radio control system for the Allen Telescope Array.
- Tested data acquisition capabilities for the ATA with the newly-installed GNU Radio system and made a successful detection of HI in the Milky Way.
- Documented the newly-developed system on a wiki and in conference proceedings.

NASA WV Space Grant Consortium, Summer Intern *May 2019-Sept. 2019*

- Refurbished a PAPER antenna at the GBO as a test instrument to support research into the HI signature from the Universe's Dark Ages.
- Assembled the antenna's hardware and digital signal processing pipelines.
- Successfully commissioned the antenna and conducted the first observations.

Green Bank Observatory, Volunteer Research Intern *June 2016-February 2022*

- Used Python to simulate the statistics of Gaussian noise in order to develop radio frequency interference (RFI) mitigation algorithms for real-time RFI excision, and developed a graphical user interface for visualizing RFI.
- Analyzed the Green Bank Telescope's pointing performance and organized the assembly of a journal article documenting the GBT's pointing and metrology systems, then revised the article based on reviewer feedback, resulting in its publication in Astronomy & Astrophysics in 2022.

Education / Public Outreach Experience

Pulsar Science Collaboratory / STARS-UP, Camp Mentor *July 2026*

- Wrote and delivered presentations on radio astronomy and general astronomy topics, and presented a live demonstration of a Small Loop Antenna DIY radio telescope.
- Chaperoned high school, undergraduate, and community college students in the Green Bank Observatory's bunkhouse facilities, taking head counts and ensuring their safety.
- Assisted with mentoring students on coding in Python as well as other astrophysical topics.
- Participated in a graduate school / career panel discussion.
- Chaperoned outdoor activities and was responsible for safely transporting students using GBO vehicles.
- Assisted with activity clean-up and other errands as needed.

Marshall University, Graduate Assistant / Adjunct Instructor *Fall 2024 – Spring 2025 / Spring 2026*

- In both Fall 2024 and Spring 2025 semesters: created and taught course material for a 4-credit hour introductory astronomy class (PS 101, Introductory Astronomy). Wrote lectures and exams based on the textbook *Astronomy 2e* by Andrew Fraknoi et al. Compiled and created lab activities utilizing resources such as Stellarium, Zooniverse, and observing with the Green Bank

20-Meter telescope, among others. Independently performed the grading for this class as well, for 20-24 students. Provided tutoring during office hours.

- Attended MU's *Critical Thinking Professional Development Workshop* in Summer 2024 to become certified to teach Critical Thinking-designated courses, such as PS 101.
- In both Fall 2024 and Spring 2025 semesters: taught General Physics I Laboratory (PHY 202) using the laboratory exercise book provided by MU's Dept. of Mathematics and Physics. Gave introductory lectures, learned to operate the lab equipment and Pasco software, answered students' questions, graded 20-24 students' completed labs and exams, and hosted office hours.
- During Fall 2025: graded labs for the Conceptual Physics Lab (PHY 101L), taught by Dr. Jon Saken. Coordinated with the instructor to ensure that my grading approach aligned with his priorities for the course.
- Received enthusiastic anonymous reviews from students at the end of each semester in both courses. Provided mentoring outside of class for multiple students.
- In Spring 2026, I revised PS 101 to be taught as two classes: a lecture class (PS 101) with a separate laboratory component (PS 101L), and revised my course material to include additional labs due to the increased lab time, and also created new assignments and activities for the lecture section to replace the time previously occupied by labs.
- Received very positive reviews from students on all classes taught.

West Virginia Science Olympiad, Optics and Astronomy Coordinator *March 2024-Present*

- Wrote exams for the WV Science Olympiad Optics and Astronomy events.
- Proctored the Optics written and lab portion and coordinated with undergraduate assistants who helped run the event in March 2024 and March 2025.
- Proctored the Astronomy exam, including preparing an online portion of the exam using JS9 software, in March 2025.
- Graded and scored the students' exams and provided rankings on the day of the event.

WV Alliance for STEM and the Arts, Co-Founder and Program Coordinator *Fall 2018-Present*

- Served as an advisor / mentor to undergraduate students who are working on the STEAM Trek educational game project.
- Assisted with maintaining the WV All STAR website.
- Attended outreach events sharing the work of WV All STAR and the STEAM Trek project with students and educators across WV and beyond.
- Assisted with writing a summary report compiling information gathered from student and teacher feedback forms to analyze the effectiveness of STEAM Trek in the classroom.
- Assisted with "The West Virginia STEM and the Arts Film Festival" in 2019.
- Chaired a workshop with the Green Bank Observatory (GBO) and a Women in STEM panel at the festival in 2019.

NASA Jet Propulsion Laboratory, Solar System Ambassador *March 2024-Present*

- Attended trainings on NASA operations, ethics, safety, and media / legislative issues.
- Organized outreach events at the Huntington Museum of Art including presentations and hands-on activities for 3rd-5th graders in Nature Camp and two public lectures for general audiences.

Small Town Universe, Documentary Subject *January 2017-Present*

- Participated in hundreds of hours of interviews / filming about radio astronomy, science education / public outreach / advocacy, and a variety of astrophysics topics ranging from studies of the early Universe to the search for extraterrestrial intelligence (SETI).
- Attended film screenings at the Cleveland International Film Festival (CIFF), the Appalachian Film Festival, the Green Bank Observatory, San Francisco DocFest, the Three Rivers Film Festival in Pittsburgh, Little Thompson Observatory in Colorado, and more.
- Participated in Q&A panels about the film at most screening events.
- Assisted with the screening tour titled: "Stories with STEAM: The Appalachian Circuit Featuring Small Town Universe." Attended and participated in Q&As at four sites in West Virginia and assisted with the setup and staffing of booths at the events to promote the nonprofit, WV All STAR, and several other STEM and the Arts initiatives in WV.

National Youth Science Camp, Directed Study Presenter *June 2020, July 2021, July 2022, June 2023*

- Wrote and presented a 3-session workshop titled: "Explore the Invisible Cosmos with Radio Astronomy" via Zoom. Taught students the basics of radio astronomy and ways to get involved as citizen scientists.

Marshall University Society of Physics Students Club *Fall 2019-Present*

- Vice President from 2020 to 2021, President in Fall 2021.
- Organized a club trip to the GBO in Fall 2019.
- Co-organized a virtual speaker series to host physicists from traditionally excluded groups.

Mapping the Milky Way Project, Mentor *April 2019*

- Served as chaperone and mentor to high school students participating in the program, and supervised their observations with the 40-Foot Telescope.

WV Science Public Outreach Team (SPOT), Ambassador *Aug. 2018-Dec. 2021*

- Completed a SPOT training course at Marshall University and became certified on the Invisible Universe 2.0 presentation.
- Attended SPOT workshops at the Green Bank Observatory.
- Gave talks on radio astronomy to hundreds of WV students.

Open Source Radio Telescopes, Co-Director *Spring 2017-Present*

- Developed an extensive curriculum and parts kits for the Small Loop Antenna project for solar flare detection, which was piloted in four schools in West Virginia.
- Provided technical assistance to students and teachers as needed.
- Assisted in the management of the organization's website and mailing list.

Get Organized for the Green Bank Observatory, Founding Member *Fall 2016-Present*

- Wrote letters to Congress and the NSF in support of the GBO in 2016, 2017, and 2025.
- Gave oral public comments at NSF hearings in Green Bank.
- Gave media interviews, and participated in two documentaries.
- Assisted with the advocacy group's mailing list, website, and social media.
- Assisted with the analysis of public comments sent to NSF to provide information about public interest in the facility to stakeholders such as members of Congress.

Community Volunteer *Fall 2015-Present*

- Ran experiments for students at GBO Family Science Day events annually.
- In Summer 2025: Assisted with events run by the Beech Fork State Park Foundation, including making baked goods for sale and volunteering at West Virginia Day and Heron Fest events.
- Volunteered at Space Race Rumpus bike races hosted at the GBO.
- Sewed masks for the WV Mask Army during COVID-19.
- Sewed joey pouches for Australian wildlife impacted by wildfires.

Cat's Eye Enterprises, LLC, Founder *Summer 2014-Present*

- Formed a small business in order to sell handmade, self-designed dolls in the likeness of great scientists for educational purposes.

Skills

- **Strong computer skills**, including proficiency in Microsoft Office products, object-oriented Python programming, Linux command-line usage, LaTeX, GNU Radio, TPOINT telescope control software, Solar System simulation packages Sorcha and ASSIST, asteroid linking algorithm heliolinx, JPL Horizons, implementation of anomaly detection algorithms using scikit-learn, Raspberry Pi, git, and WordPress. Also have experience programming in C++.
- Have over **10 years of experience in astronomy** and astronomical instrumentation research.
- Have **over 10 years of experience in mentoring, education, and public outreach** and have skills in curriculum development, teaching classes and seminars and adapting to changes on the fly, communicating complex topics to a wide variety of audiences, doing Q&A sessions, and developing lab-style activities for students of all ages.
- Experienced at **troubleshooting** and making technical decisions in time-dependent situations.
- Have **participated in well over 160 hours of observations with the GBT** and ~5 hours with the Allen Telescope Array, and attended the 2018 Single Dish Observing School at the GBO.
- **Driven and passionate** about physics, astronomy, STEAM education, outreach, and research
- Experienced at utilizing **leadership, time management, project management, collaboration (including collaboration with international teams), mentoring, and public speaking** skills**.**
- **Skilled at written and verbal communication:** consistently and clearly convey project updates to my advisors and colleagues, communicate expectations and explanations of difficult concepts to students, have experience with high-pressure media interviews, and with unscripted Q&A sessions at film screenings, public lectures, and classes.
- **Persistent** in achieving goals and finding creative solutions.
- **Resourceful, quick, and eager learner** – work well both independently and in a team.

Honors, Awards, and Memberships

- National Science Foundation Graduate Research Fellowship, awarded in April 2025
- SETI Forward Award recipient, May 2021
- Associate Member of NANOGrav, starting June 2026
- Member of the LSST Solar System Science Collaboration, starting in 2024
- Certified as a Remote Observer for the GBT (2018) and the Parkes Telescope (2020)
- Goldwater Scholar, awarded March 2020
- Speaker at Marshall University's "Bridges" TEDx event, March 2020
- Inducted into the national physics honor society, Sigma Pi Sigma

- Awarded the A. Dixon Callihan, Donald C. Martin, and Ralph Hron Memorial Physics Scholarship by the Marshall University (MU) Physics Dept. (April 2021)
- Outstanding Undergraduate Award from the MU Physics Dept. (April 2019, 2020, & 2021)
- Academic Achievement in Physics Award from the MU Physics Dept. (April 2021)
- Awarded the Alva and Dixon Callahan Scholarship by the MU Physics Dept. (April 2019, 2020, & 2021)
- Marshall University President's List (Fall 2019 – Present)
- Awarded the MU John Marshall and Honors College Scholarships (Fall 2018 – Present)
- National Merit Scholarship (Feb. 2018)

Selected Publications and Presentations

White, E. (2026). *The Link Fantastic: Detecting and Linking Simulated, Nongravitationally-Accelerating Objects With the Rubin Observatory Pipeline* (Master's thesis, Marshall University – journal article in prep).

White, E., Ghigo, F. D., Prestage, R. M., Frayer, D. T., Maddalena, R. J., Wallace, P. T., Brandt, J. J., Egan, D., Nelson, J. D., & Ray, J. (2022). Green Bank Telescope: Overview and analysis of metrology systems and pointing performance. *Astronomy & Astrophysics.*

White, E., Ghigo, F., Frayer, D., Maddalena, R., & Wallace, P. (2022). Possible new pointing models for the GBT: Definitions and comparisons. *Precision Telescope Control System (PTCS) Project Note 80, National Radio Astronomy Observatory.*

Frayer, D., White, E., & Smith, E. (2023) The GBT pointing model 6a. *Precision Telescope Control System (PTCS) Project Note 84, National Radio Astronomy Observatory.*

Jacobson-Bell, B., Croft, S., White, E., Siemion, A. P. V., Lebofsky, M., & MacMahon, D. H. E. (2025). Breakthrough Listen observations of 3I/ATLAS with the Green Bank Telescope at 1–12 GHz. *Research Notes of the American Astronomical Society*. (Accepted).

White, E., Croft, S., Pollak, A. W., & Kozel, D. (2020). GNU Radio at the Allen Telescope Array. *Proceedings of the GNU Radio Conference 5(1).*

White, E., Ghigo, F., Prestage, R., Frayer, D., Brandt, J., Maddalena, R., Nelson, J., Ray, J. (2020, June 1-3). Pointing with the 100 meter Green Bank Telescope. *Virtual poster presented at the 236th American Astronomical Society Meeting (virtual)*.

White, E., Bradley, R., Bordenave, D., & Nhan, B. (2019). Acquiring radio sky data to train machine learning-based detection algorithms for research in cosmology. *Radio Astronomy: Journal of the Society of Amateur Radio Astronomers.* Nov.–Dec. issue, pp. 29-39.

White, E. STEM Education through Radio Astronomy: Project-based learning with Small Loop Antennas. (2019, Feb.). *Poster presented at the WV Undergraduate Research Day at the Capitol, Charleston, WV, USA*.

Smith, E., White, E., Langston, G., & Prestage, R. (2018). Open Source Radio Telescopes: Astronomy projects for students, teachers, and amateurs. *Proceedings of the 2nd URSI Atlantic Radio Science Meeting.*

White, E. (2016). Statistics and visualization of radio frequency interference. *Project report, Green Bank Observatory, Green Bank, WV, USA.*

Davenport, J. R., et al. (including E. White). (2025). Technosignature Searches of Interstellar Objects. *arXiv preprint arXiv:2508.16825*.

Chandler, C. O., et al. (including E. White). (2025). NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1). *arXiv preprint arXiv:2507.13409*.